

AIMMS-MOPTA Competition 2026

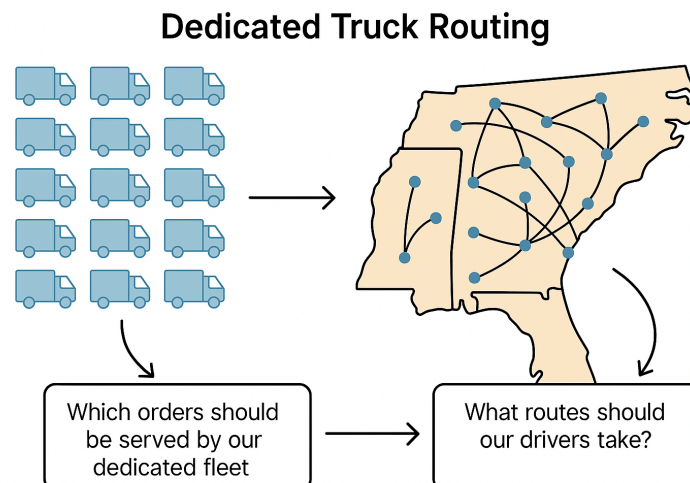
Interfor Dedicated Truck Routing

Interfor - one of North America's leading forest products companies - invites teams to tackle our **Dedicated Truck Routing** challenge: assign 20 trucks to 1,000 orders, deciding what to move with the dedicated fleet and what to tender on the open market. The goal is to balance cost and driver considerations under real operational constraints, delivering both a strong optimization model and a usable decision tool.

As one of North America's largest forest products companies, Interfor operates sawmills and distribution networks across the US and Canada, supplying customers around the globe with sustainable building materials. With an annual lumber capacity of roughly 4.7 billion board feet, our operations demand precision, reliability, and smart decisionmaking every day.

One of the puzzles we face in our US South operations is **dedicated truck routing**. Our logistics team manages a fleet of up to 30 dedicated trucks, filling in gaps in market-rate carrier availability while guaranteeing routes that leave drivers close to home at week's end, helping us meet our customer commitment.

For the AIMMS-MOPTA competition, we frame this puzzle at practical scale: **20 trucks** and **1,000 orders**. The core decision is twofold: *which orders should be served by our dedicated fleet*, and *what routes should our drivers take* all with a focus on reducing our total freight cost.



Our freight cost is calculated on a lane-by-lane basis, where each lane is an origin-destination pair of cities. Certain lanes are difficult to tender on the open market due to truck supply challenges, thus incurring an additional charge.

Some additional constraints to consider:

- Drivers must earn a minimum of \$4000 per week
- Drivers' final load must be delivered within 100 miles of their home
- Drivers must have at least 2 consecutive days off per week, and this must occur after delivery of a shipment and returning to their home (i.e., they cannot pick up a load, take 2 days off, then deliver)
- Maximum of 450 miles travelled per day by each driver
- Maximum of 50% weekly mileage can be deadhead (i.e., driving with no load)

Other useful information:

- For this problem, Carriers = Trucks in a 1-to-1 relationship
- Trucks can take a multi-stop route (i.e., Pick up from Mill A → Deliver to Customer A → Pick up from Mill B → Deliver to Customer B → etc...)
- Each shipment will take exactly 1 truck to deliver
- A shipment can arrive at any point in the week leading up to the Due Date. E.g., a shipment with a Due Date of January 24th can be delivered between January 18th and January 24th

Outcomes

Our logistics team requires an interactive planning tool using AIMMS that takes as input the order file, allows the user to modify parameters and data (e.g., minimum freight spend per week, dedicated fleet size), and returns a routing plan by carrier. The routing plan should detail which loads the carrier will transport, on which days, and the total freight cost.

Extension

We welcome any attempts to extend the model, including sourcing additional data, and offer these suggested starting points:

- Are there other costs that can be considered when making the dedicated/open-market decision?
- How does the number of dedicated drivers impact freight savings?